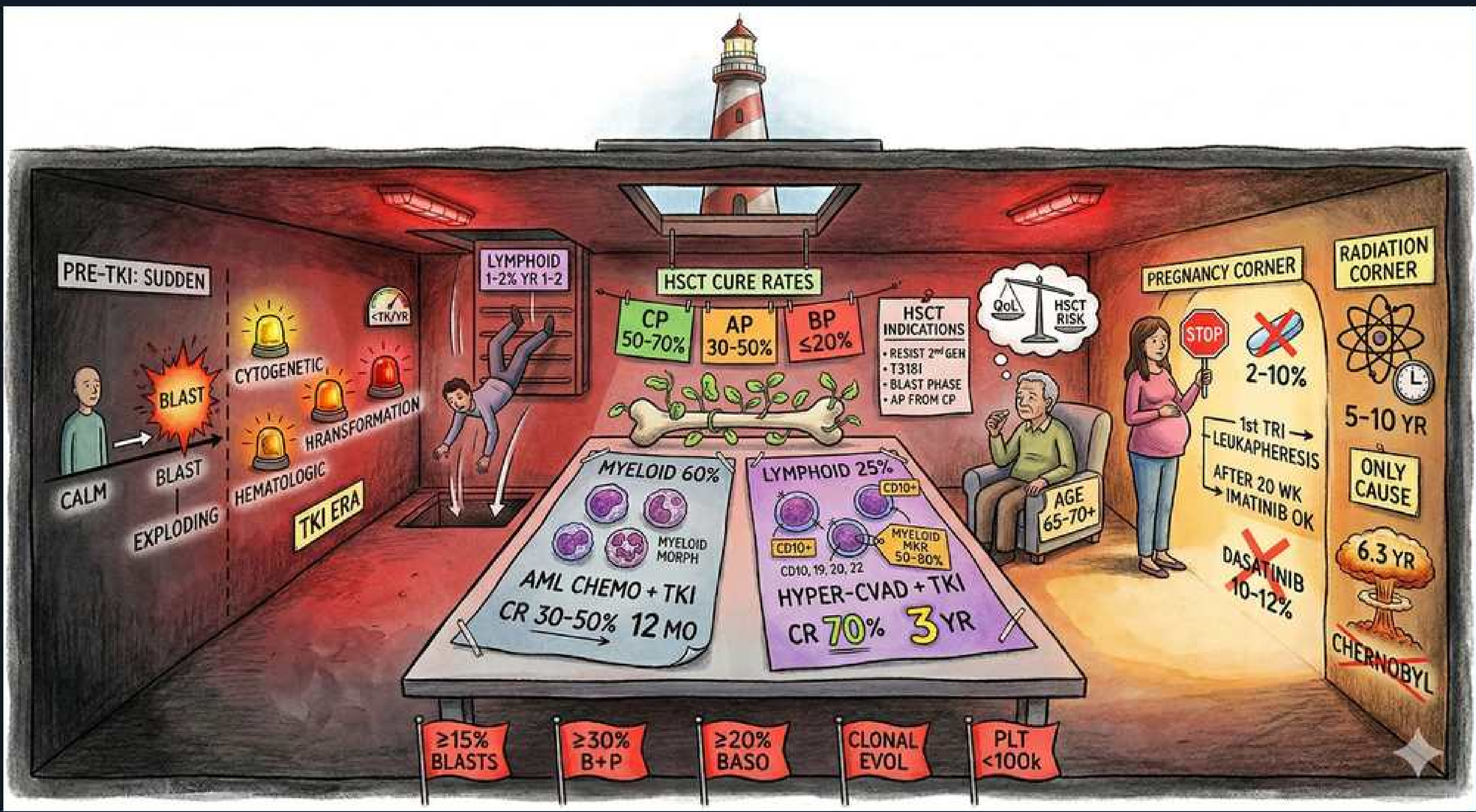
CML — Blast Crisis & Special Situations



1. Blast crisis 'exploding'

WHAT YOU SEE
PRE-TKI: SUDDEN, CALM then BLAST EXPLODING figures

WHAT IT ENCODES
In the pre-TKI era, chronic-phase CML inevitably transformed to blast crisis within ~3-5 years as additional mutations accumulated on the BCR-ABL1 clone. Transformation can be abrupt; blast crisis behaves like an acute leukemia with marrow failure, and median survival without transplant is only months.

BOARD TRAP
Sudden clinical deterioration, new cytopenias, bone pain, or extramedullary disease in a known CML patient = suspect blast crisis even if blasts seem borderline — recheck marrow and karyotype.

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2. ≥15% blasts criterion

WHAT YOU SEE
Red tag ≥15% BLASTS at bottom

WHAT IT ENCODES
Rising blasts mark progression: 10-19% defines accelerated phase and ≥20% (WHO) defines blast crisis (older ELN ≥30%). Different criteria sets use slightly different cutoffs (this drawing's ≥15%/≥30% reflect older ELN accelerated/blast thresholds).

BOARD TRAP
Know the WHO numbers as the default: accelerated 10-19%, blast crisis ≥20%. If a question quotes ≥30% for blast phase it is using older ELN criteria.

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3. ≥30% B+P

WHAT YOU SEE
Red tag ≥30% B+P

WHAT IT ENCODES
Blasts plus promyelocytes ≥30% is an accelerated/transformation criterion in older classification schemes. It captures the immature myeloid expansion that accompanies acceleration.

BOARD TRAP
Don't confuse 'blasts alone' thresholds with 'blasts + promyelocytes' thresholds — the latter is an older ELN accelerated criterion.

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4. ≥20% basophils

WHAT YOU SEE
Red tag ≥20% BASO

WHAT IT ENCODES
Peripheral basophils ≥20% is a standalone accelerated-phase criterion. Worsening basophilia tracks with disease progression and predicts inferior outcomes.

BOARD TRAP
Basophils ≥20% = accelerated phase. Don't mix this number up with blast thresholds (10-19% accel, ≥20% blast crisis).

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5. Clonal evolution

WHAT YOU SEE
Red tag CLONAL EVOL

WHAT IT ENCODES
New clonal cytogenetic abnormalities (clonal evolution) beyond the original Ph chromosome — +8, i(17q), double Ph, 3q26.2 — indicate acceleration and impending blast crisis. Detected only by full karyotype.

BOARD TRAP
Clonal evolution is invisible to FISH/PCR — only conventional karyotype catches the new abnormalities that define this accelerated-phase criterion.

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6. Platelets <100k

WHAT YOU SEE
Red tag PLT <100k

WHAT IT ENCODES
Persistent thrombocytopenia (platelets <100k) NOT attributable to therapy is an accelerated-phase criterion, reflecting marrow failure as the malignant clone crowds out normal hematopoiesis.

BOARD TRAP
ThrombocytOSIS is typical of chronic-phase CML; persistent unexplained thrombocytoPENIA signals acceleration/transformation, not just a drug side effect.

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7. Myeloid blast crisis 60%

WHAT YOU SEE
MYELOID 60% table, AML CHEMO + TKI, CR 30-50% 12 MO

WHAT IT ENCODES
Myeloid blast crisis (~60-70% of cases) is treated like AML — intensive induction chemotherapy combined with a TKI (often a 2nd/3rd-gen TKI based on mutation status), with the goal of returning to chronic phase to bridge to allogeneic HSCT. CR rates are low (~30-50%) and remissions short (median ~12 months).

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8. Lymphoid blast crisis 25%

WHAT YOU SEE
LYMPHOID 25% table, HYPER-CVAD + TKI, CR 70% 3 YR

WHAT IT ENCODES
Lymphoid blast crisis (~20-30%, usually B-lineage) is treated like Ph+ ALL — ALL-type chemo such as hyper-CVAD plus a TKI, then HSCT. CR rates (~70%) and durations (years) are markedly better than myeloid.

BOARD TRAP
Lymphoid blast crisis has a BETTER prognosis and response than myeloid blast crisis — a classic comparison question. (Distinguish lymphoid blast crisis of CML from de novo Ph+ ALL clinically/molecularly — p190 favors ALL.)

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9. HSCT cure rates by phase

WHAT YOU SEE

HSCT CURE RATES: CP 50-70%, AP 30-50%, BP ≤20%

WHAT IT ENCODES

Allogeneic HSCT is the only curative therapy, but outcome depends heavily on phase at transplant: chronic phase ~50-70% cure, accelerated ~30-50%, blast phase ≤20%. The strategy in advanced disease is to use chemo+TKI to return to chronic phase, then transplant.

BOARD TRAP

HSCT outcomes are BEST in chronic phase and worst in blast phase — so for patients who need transplant, do it earlier (less-advanced disease), not after multiple relapses.

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10. HSCT indications

WHAT YOU SEE

HSCT INDICATIONS: resist 2nd gen, T315I, blast phase, AP from CP

WHAT IT ENCODES

HSCT is reserved for: failure/resistance to ≥2nd-generation TKIs, T315I mutation when ponatinib/asciminib are unavailable or fail, blast phase, and accelerated phase arising from prior chronic phase. It is NOT used first-line in chronic phase.

BOARD TRAP

In the modern era HSCT is salvage/advanced-disease therapy, not initial treatment — TKIs are first-line for chronic phase. Reserve transplant for the listed high-risk situations.

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11. Pregnancy corner

WHAT YOU SEE

PREGNANCY CORNER, STOP pills, leukapheresis, IFN-OK

WHAT IT ENCODES

TKIs are teratogenic and contraindicated in pregnancy, especially the first trimester (imatinib embryopathy). Management of CML in pregnancy: stop the TKI, control counts with leukapheresis if needed, and use interferon-alpha (safe in pregnancy) for cytoreduction. Hydroxyurea is avoided. Some protocols cautiously resume imatinib after ~week 20.

BOARD TRAP

Do NOT give TKIs in the first trimester — they are teratogenic. Interferon-alpha and leukapheresis are the pregnancy-safe options; reserve TKIs for after delivery (or cautiously after organogenesis).

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12. Radiation/benzene risk

WHAT YOU SEE

RADIATION CORNER, atom, Chernobyl, 5-10 yr

WHAT IT ENCODES

Ionizing radiation (atomic-bomb survivors, Chernobyl, prior radiotherapy) is the only well-established environmental risk factor for CML, with a latency of ~5-10 years. Benzene exposure is more strongly linked to AML/MDS.

BOARD TRAP

Radiation is the classic CML risk factor (latency 5-10 yr). CML is generally NOT inherited and not strongly linked to most chemical exposures.

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13. Median age 65-70

WHAT YOU SEE

AGE 65-70 elderly figure

WHAT IT ENCODES

Advanced/blast-phase presentations skew older, and median CML age overall is ~55-65 (older in some registries). Age and comorbidity drive fitness assessment for intensive chemotherapy and HSCT.

BOARD TRAP

CML is rare in children; an older patient with advanced disease may not tolerate AML-type induction or transplant — fitness, not just age, guides therapy.
